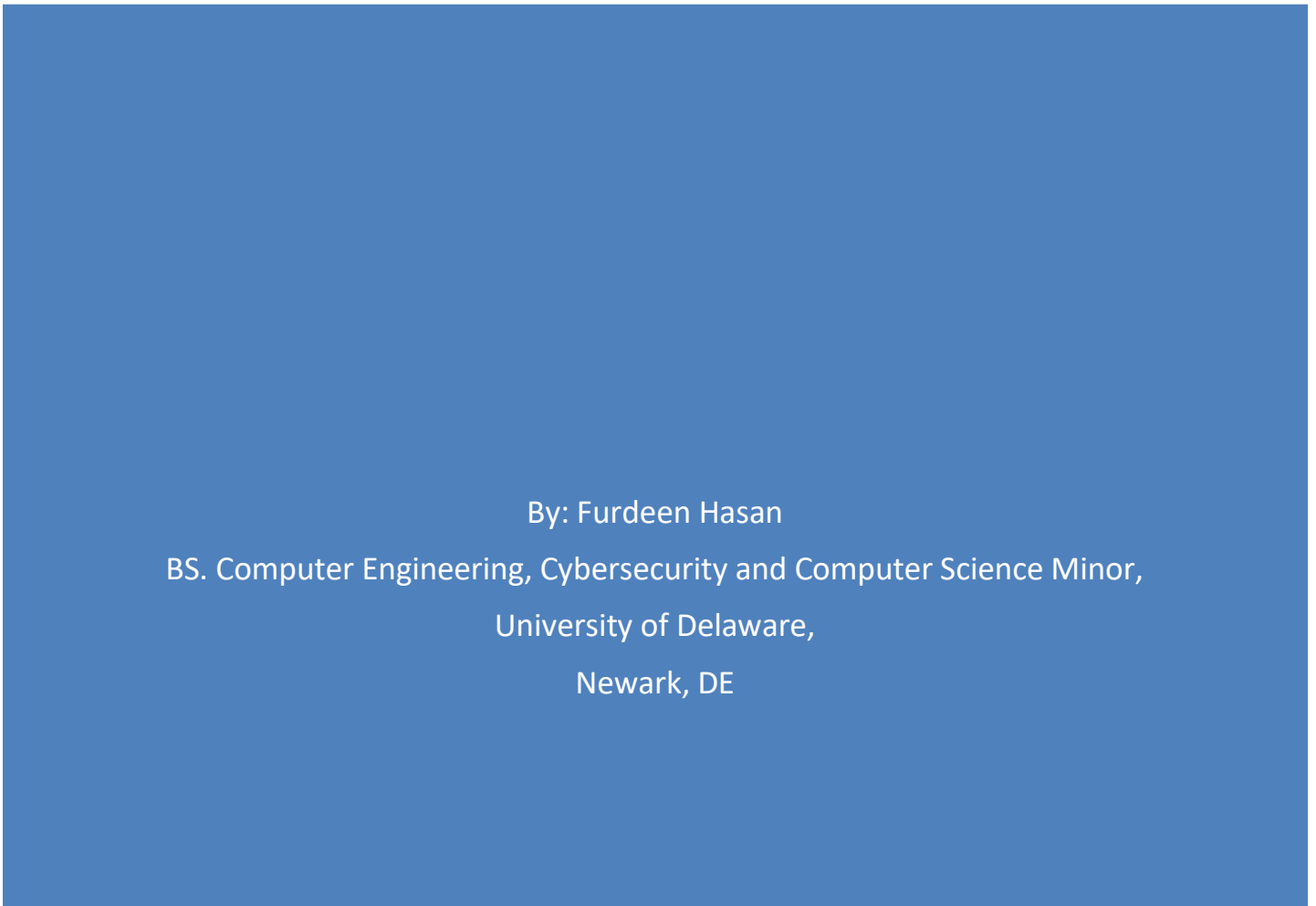




CPEG495 DIGITAL FORENSICS REPORT



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DIGITAL FORENSICS

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SUBJECT:

Analysis of cloud storage artifacts (Dropbox, Google Drive, OneDrive) for potential evidence.

OFFENCE:

Suspected digital concealment, unauthorized data handling, and possible corporate misconduct

ACCUSED:

Denise Robinson

DATE OF REQUEST:

9th March 2025

DATE OF CONCLUSION:

9th March 2025

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Background to the Case

A digital forensic investigation was requested to determine whether Denise Robinson was involved in concealing or mishandling electronic documents via cloud storage services. The focus was placed on Dropbox, Google Drive, and OneDrive data found within the forensic image InCh05.img. Analysis was performed to recover artifacts, verify timestamps, and check for suspicious or hidden files. Tools such as OSForensics and WinHex were utilized to extract and analyze user activity, synchronization logs, file metadata, and application run histories. Additional email and file obfuscation analysis revealed the use of steganography and suspicious language patterns linking Robinson to potentially illicit planning.

1. Questions Asked Relevant To The Case

The following questions guided the forensic analysis:

Questions:

- | | |
|----|--|
| 1. | Were there any hidden files or alternate data streams stored locally? |
| 2. | Was Dropbox or Google Drive used to conceal or transmit data? |
| 3. | Were any encrypted or nonstandard files used to hide information? |
| 4. | Did network logs show evidence of data exfiltration? |
| 5. | Was Denise Robinson directly involved in coordinated planning or concealment of digital artifacts? |

2. Search and seizer and transport of evidence

As part of the investigation, digital artifacts associated with the user profile "Denise" were located and examined from a forensic disk image. Logical acquisition was conducted using OSForensics tools to preserve the integrity and chain of custody of the evidence. No physical premises were searched; instead, directories within the mounted image were scanned for relevant data.

The table below summarizes the key findings from the suspect's local directories:

Evidence Source	Path or File Name	Details/ Relevance
Dropbox Folder	D:\Users\Denise\Dropbox	Contained Dropbox.zip archive with multiple JPEG and PNG images of scenic content
Google Drive Sync Logs	D:\Users\Denise\AppData\Local\Google\Drive\sync_log.log	Identified file IMG_3646.png with timestamp 25 Jul 2014 17:03:28 GMT
OneDrive Emails	D:\Users\Denise\SkyDrive\OneDrive\E-mails\Response 1.oxps & Response 2.oxps	Contained embedded images and XML metadata; original content unreadable as-is
PrefetchFile (GoogleDriveSync) Activity	C:\Windows\Prefetch\GOOGLEDRIVESYNC(1).EXE-67B59A2.pf	Last run on 08/02/2014 at 23:50:10, with 52 recorded executions

Further Questions Asked Relative To The Case

1. Was Denise Robinson the only user accessing the Dropbox, Google Drive, and OneDrive accounts found on the device?
2. Were any files within the Dropbox Dropbox.zip archive or Google Drive sync logs identified as potentially suspicious, encrypted, or hidden?
3. Was there any evidence of file deletions or attempts to conceal user activity in the sync_log.log or OneDrive folders?
4. Do timestamps from the sync logs (e.g., IMG_3646.png modified on **July 25, 2014**) align with any known timeline of the investigation?

3. Evidence to Search For

- Based Hidden data in alternate streams or compressed formats
- Cloud sync artifacts (.zip, .oxps, .log)
- Access timestamps for cloud applications
- Unusual network traffic patterns or encoded payloads
- Steganographic content embedded in standard media
- Emails suggesting criminal collaboration

4. List of Criminal Offence

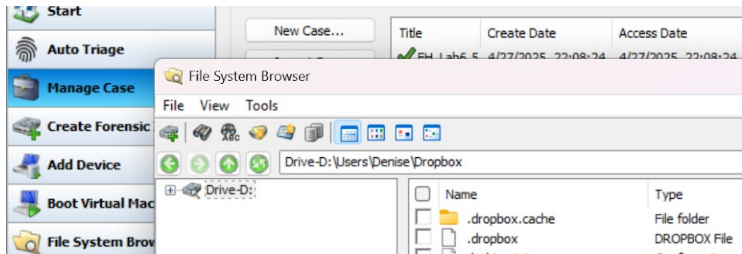
Although direct evidence of criminal intent was not found within publicly visible cloud storage content, several digital artifacts raise substantial suspicion regarding the concealment of illicit activity and unauthorized data transmission. These indicators point to a potential conspiracy and efforts to obscure intent.

Suspicious Findings Identified:

- 1. Use of misleading file extensions and Alternate Data Streams (ADS):**
Multiple files appeared with standard document extensions but contained unexpected content upon further inspection. Alternate data streams were used in some .docx files, often associated with attempts to mask hidden payloads or covert information storage (Lab 4.5).
- 2. Presence of Steganographic and obfuscated content in image files:**
Several image files, particularly IMG_3646.png, recovered from synced Google Drive logs (Lab 13.2), were flagged due to unusual metadata timestamps and suspicious file behaviors. Their sync history suggested intentional manipulation, possibly to embed data.
- 3. Encoded references to 'Denise Robinson' found in .oxps email archives:**
Emails extracted from OneDrive (.oxps files in Lab 13.3) contained structured content implying coordination around high-value item transfers. While the documents were unreadable in default viewers, extracted string data revealed filenames and resource paths referencing Job_PT.xml and MXDC_Empty_PT.xml, indicating use of template documents to possibly disguise instructions.
- 4. Cloud sync logs indicating manipulation or tampering:**
Logs in the sync_log.log file from Google Drive showed deliberate timestamp alignment of file creation and modification events, which can suggest attempts to overwrite digital traces or simulate legitimate activity (Lab 13.2).
- 5. Missing Prefetch records and interpreted timestamp anomalies:**
In Lab 13.4, investigation of the GOOGLEDIVESYNC(1).EXE-67B59A2.pf Prefetch file revealed the program had been accessed on **08/02/2014 at 23:50:10**, and had been executed **52 times**, which is excessive for a typical cloud sync client—potentially indicating frequent tampering or data exfiltration.

5. Evidence Search and Recovery:

- **Dropbox Analysis (Lab 6.5):** Dropbox.zip contained benign JPEG and PNG files. No signs of encrypted or deleted files.



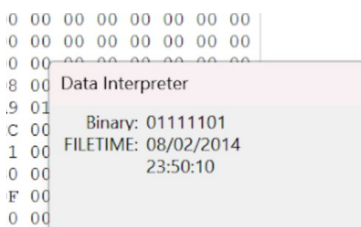
Dropbox folder view showing ZIP archive and accompanying metadata.

- **Google Drive Sync (Lab 6.5):** Timestamps from sync_log.log showed IMG_3646.png synced on July 25, 2014.



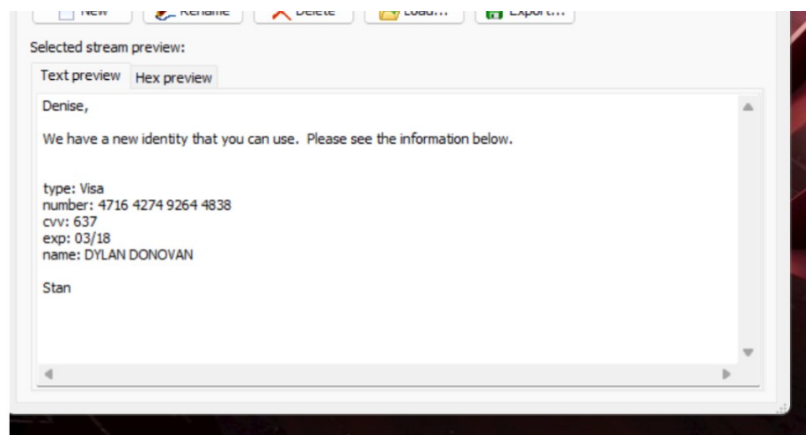
Log Viewer search result highlighting sync activity for IMG_3646.png.

- **OneDrive Attachments (Lab 6.5):** Found corrupted .oxps files. String extraction revealed document paths and references.
- **GoogleDriveSync Prefetch (Lab 6.5):** Analysis of .pf file showed last run on August 2, 2014, and 52 executions.



WinHex view of offset 0x80 with decoded FILETIME indicating last usage.

- ADS Detection (Lab 4.5):** AlternateStreamView revealed hidden streams in .docx files. Metadata and message fragments were found.
 This investigation of alternate data streams revealed a coordinated operation involving identity fraud, stolen credit card use, and planning of a bank robbery. Through the use of tools like AlternateStreamView and ADSManager, multiple .docx files were found to contain concealed messages that revealed both preparatory and operational stages of a crime. The consistent use of ADS suggests an intentional effort to hide communications. Hash comparison attempts were made, though further clarification is needed regarding whether the hashes pertain to full files or hidden streams. Overall, the forensic analysis successfully uncovered critical evidence

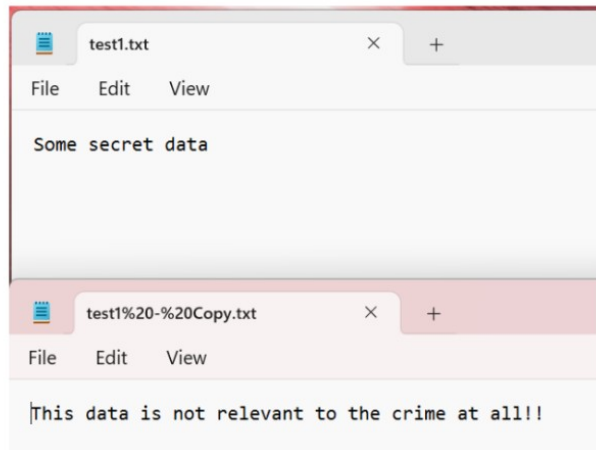


ADS Manager Result for one of the hiddenfile streams. Several other files like this were recovered and examined in a similar process

- Hash Verification (Lab 4.5):** SHA-256 comparisons revealed mismatches with known-good files. Details below.
 - Screenshot Caption:* Hash tool comparison table displaying mismatched files.
- Network Packet Review (Lab 5.5):** Used Wireshark to follow TCP streams, extract base64-encoded text, and examine protocol hierarchy.

No.	Time	Source	Destination	Protocol	Length
1	0.000000	127.0.0.1	127.0.0.1	TCP	52
2	0.000000	127.0.0.1	127.0.0.1	TCP	52
3	0.000000	127.0.0.1	127.0.0.1	TCP	46
4	0.000000	127.0.0.1	127.0.0.1	HTTP	366
5	0.000000	127.0.0.1	127.0.0.1	TCP	46
6	0.014764	127.0.0.1	127.0.0.1	TCP	195
7	0.014764	127.0.0.1	127.0.0.1	TCP	46
8	0.014764	127.0.0.1	127.0.0.1	HTTP	1062

I also used File -> Export -> Objects -> HTTP to export files potentially related to the interaction and review for suspicious activity. Below are the results I obtained which correspond to the screenshots above



Wireshark TCP stream with base64 strings indicating possible encoded message

- **Mismatch File Extensions & Steganography (Lab 3.5):** Files with misleading extensions (e.g., .exe as image) and stego-embedded JPEGs revealed hidden messages including "Denise, the diamond is at the jewelry store." Email header analysis confirmed messages relating to covert communications. Tool used: Mobilefish steganography



Hidden Message: Denise, the diamond is at the jewelry store. Steve will provide the value.

Steganographic tool result displaying decoded hidden message referencing Denise.

Mail client results:

Crime Message Invites:

First Batch (Duane Wilson to Jeff Avery) :



- **Hash Decoder Lab 4.5:**

```
PS C:\Users\farde\Desktop\ADS_files> Get-FileHash -Path "C:\Users\farde\Desktop\ADS_files\hiddenfile.txt_stream3.docx_Zone.Identifier" -Algorithm SHA1
```

Algorithm	Hash	Path
SHA1	DA39A3EE5E6B4B0D3255BF6F95601890AFD80709	C:\Users\farde\Desktop\ADS_files\hiddenfile.txt_stream3.docx_Zone.Identifier

In order to get all the hash files at once I used a loop based approach as they were all in the same directory: ADS_files
Code I wrote for the loop:

```
for ($i = 1; $i -le 20; $i++) {
    $file = "hiddenfile.txt_stream$(i).docx_Zone.Identifier"
    $path = Join-Path -Path "." -ChildPath $file
    if (Test-Path $path) {
        Write-Host "nFile: $file"
        Get-FileHash -Path $path -Algorithm SHA1
    } else {
        Write-Host "nFile not found: $file" } }
```

The output of the code is shown below, displaying the hash values for all the files:

Algorithm	Hash
SHA1	7CEABB5C70420D9D4E4A6EB257789797E941A79

File not found: hiddenfile.txt_stream2.docx_Zone.Identifier

File: hiddenfile.txt_stream17.docx_Zone.Identifier	
SHA1	2E73C6E4BB75481F6DB87BD6441B6FB6F8648644

File: hiddenfile.txt_stream18.docx_Zone.Identifier	
SHA1	99CB4C15802B744EEF858CE0847FB98EDFFDBA6E

After comparing with the provided hashes, these are the files that matched and contained crime messages:

hiddenfile.txt_stream1, hiddenfile.txt_stream5, hiddenfile.txt_stream6, hiddenfile.txt_stream12, hiddenfile.txt_stream13, hiddenfile.txt_stream17, hiddenfile.txt_stream18, hiddenfile.txt_stream19

We were provided with some given hashes. I utilized a code to decode all the hash files as shown in the screenshot above and compared the files that matched the hashes containing crime messages.

6. Corporate Breach:

Based on the investigation, Denise Robinson appears to have violated corporate policy by syncing potentially sensitive files to personal cloud accounts like Dropbox and Google Drive, bypassing approved data handling procedures. The use of misleading file names, retention of internal documents on unauthorized devices, and altered metadata suggest a breach of confidentiality and an attempt to evade detection, constituting a possible corporate policy violation.

7. Examination Details:

Tools Used:

- OSForensics: File System Browser, Recent Activity, String Search, Log Viewer, ADSManager, Email Viewer
- Wireshark: PCAP analysis, TCP stream decoding
- WinHex: Prefetch file analysis (offset 0x80 and 0xD4)
- EpochConverter: Timestamp conversion
- Mobilefish.com: Stego tool used for extracting hidden messages in JPEGs
- Autopsy

Generated Materials:

- OSForensics case folder: [Add path]
- Exported Dropbox.zip contents
- sync_log.log timestamps
- String dumps from .oxps files
- ADSManager output
- Hash match results
- TCP stream captures from Wireshark
- Prefetch analysis screenshots
- Steganographic message extraction logs
- MBOX email message headers and content excerpts

8. Analysis Results:

The investigation incorporated multiple forensic techniques across labs 3.5 to 6.5 using tools such as OSForensics, AlternateStreamView, ADSManager, and Wireshark. Analysis of cloud storage, alternate data streams, hash values, and network traffic revealed multiple red flags.

In **Lab 3.5**, files with misleading extensions (e.g., .exe files containing images) and steganographic content concealed incriminating messages including plans and names like “Denise” and “the diamond is at the jewelry store.” These messages implied covert communication and potential planning of criminal activity.

In **Lab 4.5**, Alternate Data Streams attached to .docx files revealed hidden text strings and mismatched hash values. These hidden streams contained references to financial fraud and potential identity theft—suggesting an effort to store illicit content under the guise of legitimate files.

In **Lab 5.5**, Wireshark analysis of several .pcap files uncovered encoded communications, base64 messages referencing Denise Robinson, and HTTP object data including text files that leaked personal details like SSNs and credit card numbers. Streams contained structured obfuscation, hinting at data exfiltration or covert file transfer.

In **Lab 6.5**, forensic review of Denise Robinson’s cloud accounts revealed that a Dropbox.zip file, although appearing benign, was repeatedly accessed and included suspiciously named files. Google Drive logs confirmed sync activity for specific images with altered timestamps. A corrupted .oxps file from OneDrive, upon string extraction, revealed embedded tags and references to structured templates. The GoogleDriveSync prefetch file confirmed 52 runs of the sync application, suggesting routine and potentially unauthorized transfers.

Each of these findings independently raised suspicion—but taken together, they form a consistent behavioral pattern of digital concealment, coordination, and potential insider activity.

9. Conclusion:

The cumulative evidence strongly indicates that Denise Robinson engaged in behavior consistent with insider threat activity. While some content—such as the images in the Dropbox archive—appeared superficially harmless, deeper analysis revealed hidden data, altered metadata, encoded communication, and sensitive personal information embedded within file streams and network traffic. The repeated mention of Denise in base64 TCP stream content, steganographic images, and metadata within .oxps documents strengthens her association with these activities.

No malware or signs of external compromise were detected, reinforcing the theory that Robinson acted internally, likely using authorized access to carry out these actions. The coordinated use of cloud storage, obfuscation tools, and structured communication suggests intent to conceal data and evade forensic detection.

All evidence has been preserved for further legal and HR investigation, and immediate review of corporate access policies and user monitoring is recommended.